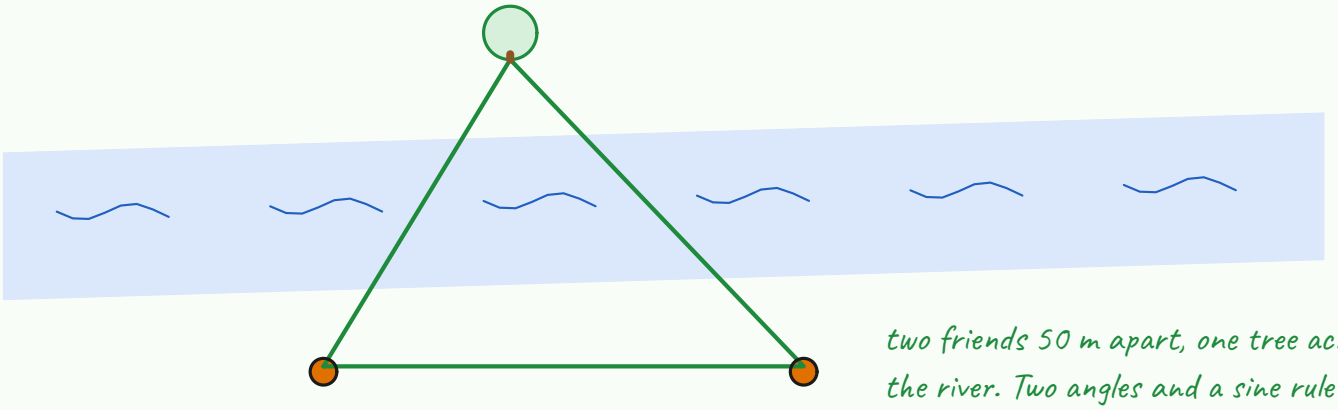
Chapter 31

Solving any triangle

measure a river without getting wet



two friends 50 m apart, one tree across the river. Two angles and a sine rule give the width.

SOHCAHTOA only works when there's a right angle. Real triangles (plots of land, roof trusses, navigation) usually don't have one. This chapter gives you two tools that work on **any** triangle: the **sine rule** and the **cosine rule**, plus a map for choosing between them.

Goes with **Lecture 6** (triangle types and notation, sine rule, cosine rule, the clock example, answering questions).

△ Names, and the 180° rule

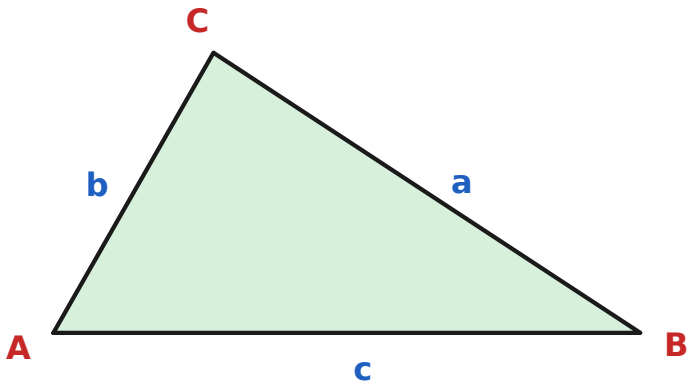
RULE: triangle notation and the angle sum

Angles are capital letters A, B, C; the side **opposite** each angle is the same letter in lower case: side a faces angle A.

The three angles add up to $180°$: $A + B + C = 180°$. Two angles always give the third.

The biggest angle faces the longest side; the smallest angle faces the shortest side.

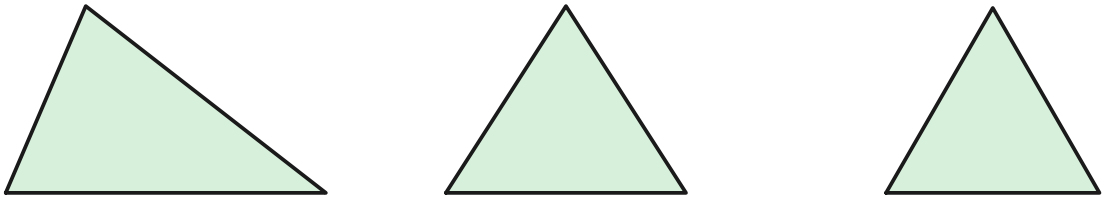
Lecture 6's labelling: **capital letters for the angles** (at the corners) and the **same letter in lower case for the side opposite**. Side a faces angle A:



a is opposite A,
b is opposite B,
c is opposite C.

Bigger angle ↔ longer side opposite it.

Three kinds (Lecture 6): **scalene** (all sides different, from Greek "unequal"), **isosceles** (two equal sides: "equal legs") and **equilateral** (all three equal, so all angles 60°).



scalene **isosceles** **equilateral**

Angles in a triangle add up to 180°. Tear the three corners off a paper triangle and line them up: they always make a straight line. So two angles always give you the third.

△ The sine rule: drop a height

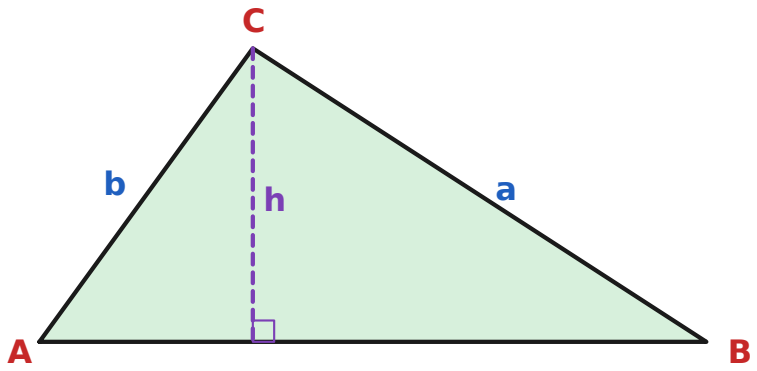
RULE: the sine rule

In **any** triangle: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$. Each side over the sine of the angle **opposite** it.

To find an **angle**, flip it: $\frac{\sin A}{a} = \frac{\sin B}{b}$ (the unknown goes on top).

Use it when you know a **matching pair** (a side and the angle opposite it) plus one more side or angle. Use only the two fractions you need.

Where does it come from? Drop a line straight down from C to the base, height h. That splits the triangle into **two right-angled triangles**, where SOHCAHTOA works again:



left triangle: sin A = h / b
so h = b sin A

right triangle: sin B = h / a
so h = a sin B

Both expressions are the same height, so

$$b \sin A = a \sin B$$

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$ ← *÷ both sides by sin A sin B*

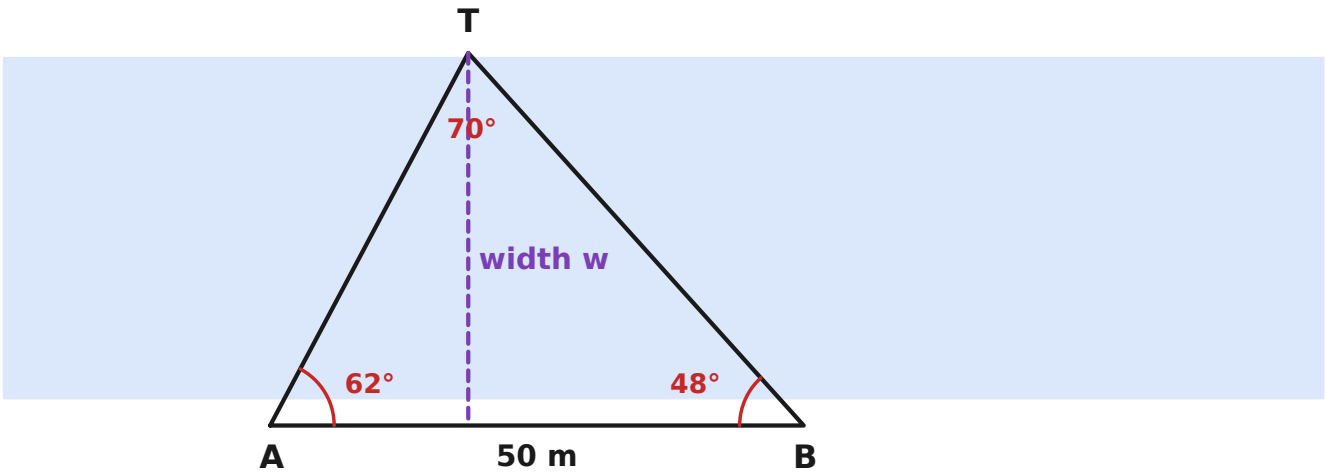
Dropping the height from a different corner brings in c the same way. The **sine rule**:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$ ← *each side ÷ the sine of its opposite angle*

When to use it: you know a **matching pair** (a side and the angle opposite it) plus one more fact. Use only the two fractions you need.

△ Measuring the river

Ana (A) and Ben (B) stand 50 m apart on one bank. Each measures the angle between the bank and a tree T on the far side: 62° at A, 48° at B. How wide is the river?



$$T = 180^\circ - 62^\circ - 48^\circ = 70^\circ$$ ← angle sum: now 50 m faces a known angle

$$\frac{AT}{\sin 48^\circ} = \frac{50}{\sin 70^\circ}$$ ← sine rule: AT faces B, 50 faces T

$$AT = \frac{50 \sin 48^\circ}{\sin 70^\circ} = 39.54 \text{ m}$$ ← × sin 48°

$$w = AT \sin 62^\circ = 34.9 \text{ m}$$ ← SOH in the right triangle at A

Lecture 6's example: $a = 17$, $b = 2$, $A = 24^\circ$. Find B:

$$\frac{17}{\sin 24^\circ} = \frac{2}{\sin B} \Rightarrow \sin B = \frac{2 \sin 24^\circ}{17}$$

$$B = \sin^{-1}\left(\frac{2 \sin 24^\circ}{17}\right) = 2.743^\circ$$

△ The cosine rule: Pythagoras plus a correction

RULE: the cosine rule (all three forms)

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$b^2 = a^2 + c^2 - 2ac\cos B$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

Pattern: the side on the left is **opposite the angle** in the cosine; the other two sides appear squared and in the $2 \times \ldots$ term.

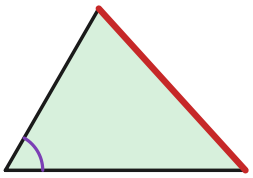
Angle form (rearranged): $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$, $\cos B = \dfrac{a^2 + c^2 - b^2}{2ac}$, $\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$.

Use it with **two sides and the angle between them** (find the third side) or **all three sides** (find an angle).

If you know **two sides and the angle between them**, there's no matching pair, so the sine rule is stuck. The **cosine rule** takes over:

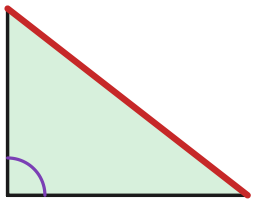
$$a^2 = b^2 + c^2 - 2bc\cos A$$

Look closely: it's **Pythagoras** ($a^2 = b^2 + c^2$) with a correction term. The correction depends on how far A is from 90°:



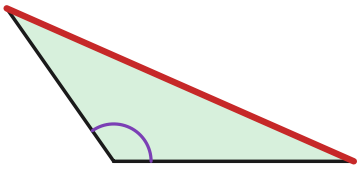
A < 90°

cos A > 0: subtract, a is shorter



A = 90°

cos A = 0: plain Pythagoras



A > 90°

cos A < 0: add, a is longer

Lecture 6's clock: hands 10 cm and 7 cm long. How far apart are the tips at 2 o'clock?

$$A = \frac{2}{12} \times 360^\circ = 60^\circ$$ ← *the angle between the hands (Chapter 27)*

$$a^2 = 10^2 + 7^2 - 2 \times 10 \times 7\cos 60^\circ$$

$$a^2 = 100 + 49 - 140 \times 0.5 = 79$$

$$a = \sqrt{79} = 8.89 \text{ cm}$$

Same rule, other letters: $a = 8$, $c = 5$, $B = 110°$. Find b (it faces B, so use the b^2 form):

$$b^2 = 8^2 + 5^2 - 2 \times 8 \times 5 \cos 110° = 89 + 27.36 = 116.36$$

cos 110° < 0: the term is added

$$b = \sqrt{116.36} = 10.79$$

Finding an angle from three sides: rearrange the rule (Chapter 14) to

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\text{sides } 5, 6, 7: \quad \cos C = \frac{25 + 36 - 49}{2 \times 5 \times 6} = 0.2, \quad C = 78.46°$$

C faces the 7

△ Which rule? Ask "what do I know?"

RULE: sine rule or cosine rule?

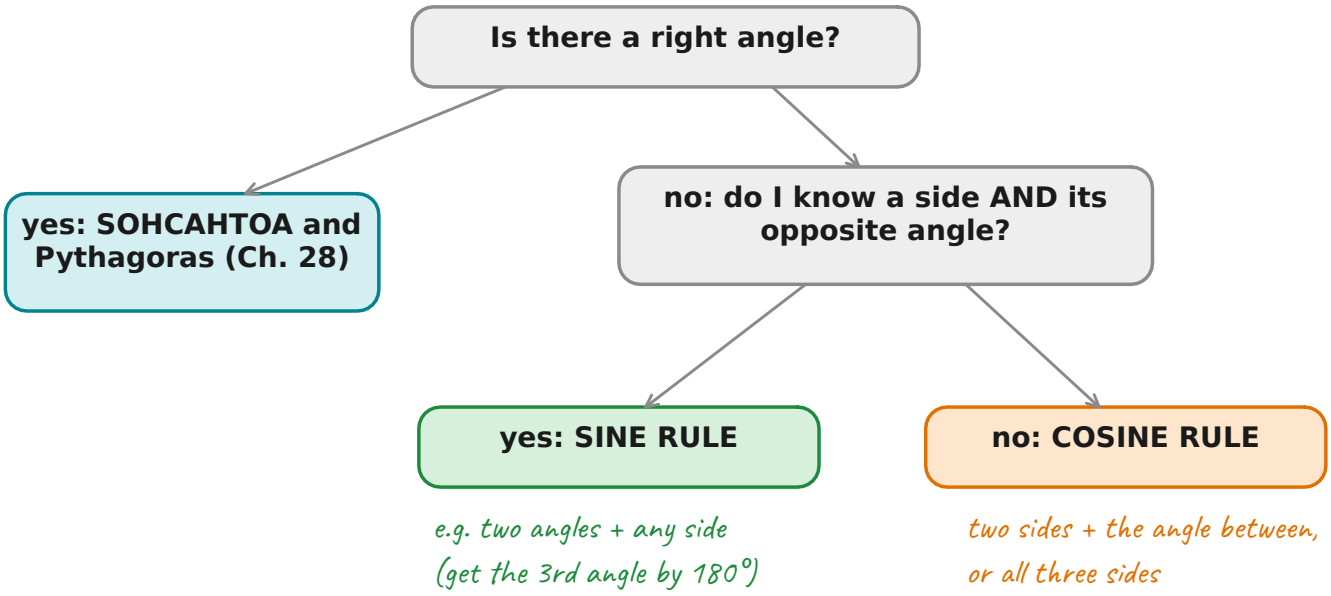
Right angle in the triangle → SOHCAHTOA and Pythagoras (Chapter 28).

Know a side **and the angle opposite it** (a matching pair) → **sine rule**.

Know two sides and the angle **between** them (SAS), or **all three sides** (SSS) → **cosine rule**.

You know	Use	Then
two angles + any side (AAS / ASA)	sine rule	third angle $= 180° -$ the other two, first
two sides + an angle opposite one of them (SSA)	sine rule (for an angle)	check the **ambiguous case** (below)
two sides + the angle between them (SAS)	cosine rule	third side; then the sine rule for an angle
three sides (SSS)	cosine rule, angle form	find the **largest** angle first
three angles only	can't be solved	only the shape is fixed, not the size

Lecture 6's advice: **first draw the triangle and fill in everything you know**. Then:



RULE: the ambiguous case (two possible triangles)

It can only happen with **two sides and a non-included angle** (SSA): you know a, b and A.

It happens only when the given angle A is **acute** and the side opposite it is the **shorter** one ($a < b$), but still longer than the height $b \sin A$.

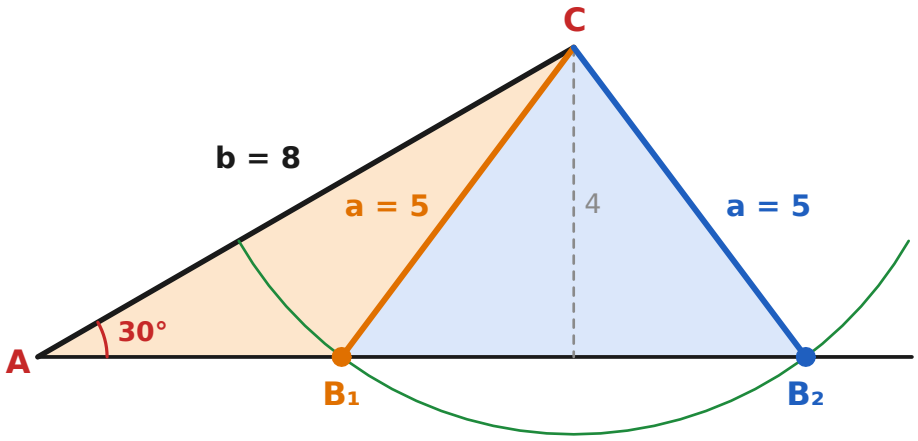
Then $\sin B$ gives two angles: $B_1 = \sin^{-1}(\dots)$ (acute) and $B_2 = 180° - B_1$ (obtuse). Keep B_2 only if $A + B_2 < 180°$.

If A is **obtuse** (or $90°$), or $a \geq b$: only **one** triangle. If $a < b \sin A$: no triangle at all (the calculator gives an error).

See it with a hinge. $A = 30°$, $b = 8$, $a = 5$. Draw the 30° angle at A and the 8-long side up to C. Side a (5 long) hangs from C on a hinge, and must reach the bottom line to close the triangle.

✎ **YOUR GO (pencil first)**

Swing the 5-long side from C down to the bottom line (a compass set to 5 does it). How many places can it land?



Green arc: everywhere the hinged side can reach. It crosses the line TWICE.

Orange triangle: A, B_1, C. Blue triangle: A, B_2, C (the orange one sits inside it).

Two different triangles, both with $A = 30°$, $b = 8$, $a = 5$. The sine rule can't tell them apart:

$$\sin B = \frac{8 \sin 30°}{5} = 0.8$$ ← *sine rule, rearranged*

$$B = 53.13° \text{ (calculator)} \quad \text{or} \quad B = 180° - 53.13° = 126.87°$$

$\sin^{-1}$ **only ever gives the acute one** (the blue triangle, at B_2). The obtuse partner (orange, at B_1) has exactly the same sine, and it fits too: $30° + 126.87° = 156.87°$, under 180°. When the hinge is **shorter than b but longer than the height** ($4 < a < 8$ here), expect two answers. Longer than b: only one. Shorter than the height: it can't reach, so no triangle.

△ Area of any triangle

RULE: area from two sides and the included angle

$\text{Area} = \frac{1}{2}ab\sin C = \frac{1}{2}bc\sin A = \frac{1}{2}ac\sin B$: half × two sides × the sine of the angle **between** them.

Find an angle from the area: $\sin C = \dfrac{2 \times \text{Area}}{ab}$. Two angles have that sine, C and $180° - C$; both can be valid.

Find a side from the area: $a = \dfrac{2 \times \text{Area}}{b\sin C}$.

"Half base × height" needs the height. But the height is $b\sin A$ (from the sine-rule picture), so with two sides and the angle between them:

$$\text{Area} = \frac{1}{2}\,bc\sin A$$

$$\text{clock triangle: } \frac{1}{2} \times 10 \times 7 \times \sin 60° = 30.3 \text{ cm}^2$$

And the **perimeter** is just the three sides added: for the clock triangle $10 + 7 + 8.89 = 25.89$ cm.

△ Find the unknown: working backwards and chaining

PROCEDURE: partial-information triangle problems

1. Sketch the triangle and label **everything** you know, including the area if given.
2. Write the equation that links what you know to what you want (sine rule, cosine rule, area formula, or $A + B + C = 180°$).
3. Solve for the unknown. For an angle from a sine, check whether $180° -$ it also fits.
4. If the rule you need has a gap, fill it first with another rule (a chain: angle → side → area).

Area → side: a triangle has area 20 cm², $b = 8$ cm and $C = 30°$. Find a.

$$20 = \tfrac{1}{2} \times a \times 8 \times \sin 30° = 2a \;\Rightarrow\; a = 10 \text{ cm}$$

½ × 8 × 0.5 = 2

Area → angle: area 30 cm², sides 8 cm and 10 cm. Find the angle C between them.

$$30 = \tfrac{1}{2} \times 8 \times 10 \sin C \;\Rightarrow\; \sin C = \frac{30}{40} = 0.75$$

$$C = 48.59° \quad \text{or} \quad C = 180° - 48.59° = 131.41°$$

both fit: two triangles

Chain (angle → side → area): $A = 50°$, $B = 60°$, $c = 12$. Find the area.

$$C = 180° - 50° - 60° = 70°$$ ← angle sum: now c = 12 has its matching angle

$$a = \frac{12 \sin 50°}{\sin 70°} = 9.78$$ ← sine rule

$$\text{Area} = \tfrac{1}{2} ac \sin B = \tfrac{1}{2} \times 9.78 \times 12 \times \sin 60° = 50.83$$

B is between a and c

✗ Common mistakes

✗ $\text{Area} = \frac{1}{2}ab\sin A$ ✓ $\text{Area} = \frac{1}{2}ab\sin C$

the angle must be the one between the two sides you use

✗ $c^2 = \ldots - 2ab\cos A$ ✓ $c^2 = \ldots - 2ab\cos C$

the side on the left faces the angle in the cosine

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Notation	side a is opposite angle A, and so on
Angle sum	$A + B + C = 180^\circ$
Sine rule	$\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$ for an angle: $\frac{\sin A}{a} = \frac{\sin B}{b}$
Cosine rule	$a^2 = b^2 + c^2 - 2bc\cos A$ also $b^2 = a^2 + c^2 - 2ac\cos B$, $c^2 = a^2 + b^2 - 2ab\cos C$
Cosine rule (angle)	$\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$ SSS: find the largest angle first
Which rule?	matching side + angle → sine rule; SAS or SSS → cosine rule
Ambiguous case	SSA with the given angle acute and the side opposite it shorter: try B and $180^\circ - B$. Obtuse given angle: one triangle only
Area	$\text{Area} = \frac{1}{2}ab\sin C$ two sides and the angle between them
Area backwards	$\sin C = \dfrac{2 \times \text{Area}}{ab}$, $\quad a = \dfrac{2 \times \text{Area}}{b\sin C}$ an angle from a sine may have two answers

△ Your turn

Draw each triangle first. Answers in grey.

Sine rule

1. A = 40°, B = 75°, a = 8. Find b.
2. a = 12, A = 110°, b = 9. Find B.
3. Two angles are 35° and 65°; the side between them is 20. Find the side opposite 65°.

1. $8 \sin 75°/\sin 40° \approx 12.02$ 2. $\sin^{-1}(9 \sin 110°/12) \approx 44.8°$ 3. third angle 80°; $20 \sin 65°/\sin 80° \approx 18.41$

Cosine rule

4. b = 5, c = 8, A = 50°. Find a.
5. Sides 4, 7, 9: find the largest angle.
6. Clock hands 10 cm and 7 cm at 4 o'clock: distance between the tips?

4. ≈ 6.13 5. $\cos^{-1}((16 + 49 - 81)/56) \approx 106.6°$ 6. 120°: $\sqrt{(149 + 70)} \approx 14.80$ cm

Choose

7. You know all three angles. Can you find the sides?
8. Area with sides 6, 9 and 30° between

7. no, only their ratios (the triangle could be any size) 8. 13.5

Find the unknown

9. A triangle has area 24 cm² and sides 8 cm and 10 cm. Find the angle between them (two answers).
10. Area 35 cm², C = 150°, a = 10 cm. Find b.
11. A = 40°, B = 70°, c = 15. Find a, then the area.
12. Sides 7 and 9 with 60° between them: find the third side, then the area.
13. A = 120°, a = 9, b = 5. How many triangles? Find B.

9. $\sin C = 48/80 = 0.6$, C = 36.87° or 143.13° 10. $35 = ½ \times 10 \times b \times 0.5$, so b = 14 cm 11. C = 70°; $a = 15 \sin 40°/\sin 70° \approx 10.26$; area = $½ \times 10.26 \times 15 \times \sin 70° \approx 72.31$ 12. $c^2 = 49 + 81 - 63 = 67$, $c \approx 8.19$; area = $½ \times 7 \times 9 \times \sin 60° \approx 27.28$ 13. A is obtuse, so one triangle: $\sin B = 5 \sin 120°/9 \approx 0.481$, B ≈ 28.76°